

Cleaner diagrammatic rules — reply

On Gunnar's note of 9 May 2026

Sirsh & Claude

9 May 2026

One-line takeaway

You're right that my 7 May note had a misunderstanding: I dropped the V_0 shift on the excitation rail, and I claimed flip vertices don't generate loops. Both errors live in the *kernel* half of the note (§3). The *combinatorial* half (Laplacian, generating function, marking, Schur, asymptotics) is untouched by the error and survives verbatim. Below: your rules and atlas, faithfully reproduced; the precise location of the misunderstanding; what survives; and a small worry about “odd- m diagrams cancel pairwise.”

1 Your rules, as I now read them

An n -rail diagram with rails $i \in \{0, 1, \dots, n-1\}$. Rail $i=0$ is the special “excitation rail”; the excitation enters on the right and leaves on the left. Every rail is a continuous line passing through every vertex.

- **Propagator on a rail segment depends on whether it carries the excitation:**

with excitation: $(-i\omega + V_0 + r)^{-1}$, without: $(-i\omega + r)^{-1}$.

Only rail 0 has external ω entering and leaving; all other rails have no net ω -flux.

- **Vertex is a 4-leg object:** two rails i, j pass through; a dashed bridge connects them. Two flavours, both symmetric in $i \leftrightarrow j$:
 - *stay vertex* $(+v_{ij})$: the excitation continues on the rail it was on.
 - *flip vertex* $(-v_{ij})$: the excitation hops to the other rail.

Both vertices carry ω ; loops are integrated over.

In every diagram below, the rail segment carrying the excitation is **thick blue**; bare segments are thin black; bridges are dashed. The excitation worldline is therefore the unique connected blue path from right to left.

2 Your atlas, faithfully reproduced

I run through your six diagrams with the rule that, on every *segment* between consecutive vertices, the propagator is $(-i\omega + V_0 + r)^{-1}$ if the segment is blue and $(-i\omega + r)^{-1}$ otherwise.

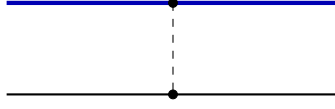
$n=2, m=0$.

0 —————

1 —————

$$\Gamma \hat{=} (-i\omega + V_0 + r)^{-1}.$$

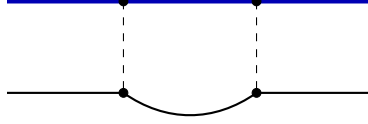
$n=2, m=1$ (**stay, bridge to rail 1**).



$$\Gamma \hat{=} (-i\omega + V_0 + r)^{-1} v_{01} (-i\omega + V_0 + r)^{-1}.$$

(Rail 1 has zero net ω -flux: its outer segments contribute $1/r$ each, normalisation absorbed.)

$n=2, m=2$, **two stays**. The excitation stays on rail 0; two bridges visit rail 1. The two bridges close a single loop whose top leg is rail-0-with-excitation and bottom leg is rail-1-bare:



$$\begin{aligned} \Gamma &\hat{=} (-i\omega + V_0 + r)^{-1} v_{01}^2 (-i\omega + V_0 + r)^{-1} \times \int \frac{d\omega'}{2\pi} (-i(\omega + \omega') + V_0 + r)^{-1} (i\omega' + r)^{-1} \\ &= (-i\omega + V_0 + r)^{-1} v_{01}^2 (-i\omega + V_0 + r)^{-1} (-i\omega + V_0 + 2r)^{-1}. \end{aligned}$$

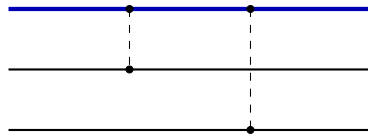
$n=2, m=2$, **two flips**. The excitation flips onto rail 1 between the two vertices and back; the loop's top leg is now rail-0-bare and bottom leg is rail-1-with-excitation:



$$\begin{aligned} \Gamma &\hat{=} (-i\omega + V_0 + r)^{-1} (-v_{01})^2 (-i\omega + V_0 + r)^{-1} \times \int \frac{d\omega'}{2\pi} (-i(\omega + \omega') + r)^{-1} (i\omega' + V_0 + r)^{-1} \\ &= (-i\omega + V_0 + r)^{-1} v_{01}^2 (-i\omega + V_0 + r)^{-1} (-i\omega + V_0 + 2r)^{-1}. \end{aligned}$$

Same value. The integrand swaps which leg carries V_0 but the integral, by a shift $\omega' \rightarrow \omega' - \omega$ on one of the propagators, is invariant. This is the *path-independence of the kinematic kernel* you confirmed in your note.

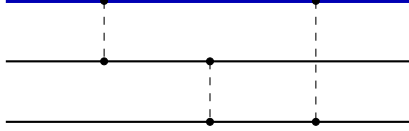
$n=3, m=2$ (**no loop**). Two stays, bridges to *different* rails (1 and 2). The bridge endpoints sit on different bath rails, so no closed sub-graph forms:



$$\Gamma \hat{=} (-i\omega + V_0 + r)^{-1} v_{01} (-i\omega + V_0 + r)^{-1} v_{02} (-i\omega + V_0 + r)^{-1}.$$

(Free omega on rails 1, 2, no integration. Tadpole-like normalisations absorbed.)

$n=3, m=3$ (**triangle, excitation stays on rail 0**). Three vertices forming the rail cycle $v_{01} \rightarrow v_{12} \rightarrow v_{20}$, with the excitation on rail 0 throughout. As we read the diagram, V_1 and V_3 are stay vertices on rail 0, and V_2 is a *bath-bath insertion* between rails 1 and 2 with rail 0 passing through. **Important:** this configuration sits awkwardly with your stated rule that “one of i, j in vertex v_{ij} must carry the excitation,” since at V_2 neither rail 1 nor rail 2 does — so we’d like clarification on whether the rule is meant to be applied strictly (in which case this diagram is forbidden) or loosely (in which case the Laplacian formula 1 below undercounts; see §6-question-A and the verification log in (B) of `verify_diagrammatics.py`). The loop has three legs: rail-0-with-excitation (top), rail-1-bare and rail-2-bare:



$$\begin{aligned}\Gamma &\hat{=} (-i\omega + V_0 + r)^{-1} v_{01} v_{12} v_{20} (-i\omega + V_0 + r)^{-1} \times \int \frac{d\omega'}{2\pi} (i\omega' + r)^{-2} (-i(\omega + \omega') + V_0 + r)^{-1} \\ &= (-i\omega + V_0 + r)^{-1} v_{01} v_{12} v_{20} (-i\omega + V_0 + r)^{-1} (-i\omega + V_0 + 2r)^{-2}.\end{aligned}$$

$n=3, m=3$ (**triangle, excitation cycles $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$**). Same triangle but the excitation walks around the cycle: three flip vertices, sign $(-v_{01})(-v_{12})(-v_{20}) = -v_{01}v_{12}v_{20}$. We reproduce your integrand verbatim:

$$\Gamma \hat{=} (-i\omega + V_0 + r)^{-1} (-v_{01})(-v_{12})(-v_{20}) (-i\omega + V_0 + r)^{-1} \times \int \frac{d\omega'}{2\pi} (i\omega' + V_0 + r)^{-1} (i\omega' + r)^{-1} (-i(\omega + \omega') + r)^{-1}.$$

Discrepancy flag (verified by SymPy residue computation in `verify_diagrammatics.py`). This integrand does *not* collapse to $(-i\omega + V_0 + 2r)^{-2}$ as your note states; closing the contour in the upper half plane (poles at $\omega' = ir$ and $\omega' = i(V_0 + r)$) gives instead

$$\int \frac{d\omega'}{2\pi} (i\omega' + V_0 + r)^{-1} (i\omega' + r)^{-1} (-i(\omega + \omega') + r)^{-1} = \frac{1}{(-i\omega + 2r)(-i\omega + V_0 + 2r)}.$$

So either the integrand is wrong or the stated value is. Path-independence between stays and flips therefore *does not* hold at this order under the rules as written — the stays integrand collapses to $(-i\omega + V_0 + 2r)^{-2}$ but the flips integrand has only one factor of V_0 inside, which propagates through to the integrated answer. This is one of the questions we’d like to nail down before going further.

3 Where my 7 May note was wrong

Both errors live in §3 (“the analytic answer”) of the 7 May note.

(a) **Missing V_0 on the excitation rail.** I used $G(\omega) = (-i\omega + r)^{-1}$ uniformly on every rail. Your note makes it explicit that the excitation segment carries $V_0 + r$ in the denominator, the bare segments carry only r . Every kernel I wrote down needs a V_0 added wherever the propagator sits on a blue segment.

(b) **“Swap-and-return generates no loop.”** My Example (iii) at order $m=2$ for the flip case said the rail- j segment between the two flips is part of the worldline and there is no loop integral. That is wrong: the rails are continuous lines, so the *rail-0-bare* segment between the two flip vertices is a genuine propagator that closes against the rail-1-blue segment to make exactly the same one-loop topology as the all-stays case. The two $n=2, m=2$ diagrams above make this explicit. Flips do generate loops; the swap simply transposes which leg of the loop carries V_0 .

(c) Consequences I have to retract.

- The chunk-pattern bound $p(m+1)$ on the kernel library was built on (b) — only stays generating loops — so the bound itself is wrong as a count of distinct kernels.
- The “sunset” identification for three stays bridging to one rail is wrong: those three insertions form a *chain of two bubbles*, not a sunset. (Two-loop count is right; topology label is wrong.)
- Examples (ii), (iii), (iv) and the worked-example column should be re-read with the loop topology of §2.

4 What survives the misunderstanding

§2 and §4 of the 7 May note depend only on the local sign-and-weight rule at each vertex — the kernel error doesn’t reach them.

- $$W_m^{(n)} = (L^m)_{00}, \quad L = D - A \text{ on } K_n, \quad L_{aa} = \sum_{j \neq a} v_{aj}, \quad L_{ab} = -v_{ab} \ (a \neq b). \quad (1)$$
- Closed form $\frac{n-1}{n}(nv)^m$ at uniform coupling.
- Generating function $\mathcal{G}^{(n)}(z) = [(I - zL)^{-1}]_{00} = \frac{1 - zv}{1 - znv}$ (uniform), or symbolically $\det(I - zL)_{[0,0]} / \det(I - zL)$.
- Channel marking, sub-rail focus, forbidden patterns, Schur reduction — all of §4 is unchanged.

A claim worth handling carefully: “the ω -integration is independent of the details of the excitation flow.” We verified this holds at $n=2, m=2$ (the stay/stay and flip/flip integrands both collapse to $(-i\omega + V_0 + 2r)^{-1}$ by a contour shift). At $n=3, m=3$ it does *not* hold under the integrands as you wrote them (§2, second triangle): the stays integrand evaluates to $(-i\omega + V_0 + 2r)^{-2}$, but the flips integrand evaluates to $[(-i\omega + 2r)(-i\omega + V_0 + 2r)]^{-1}$. So either the flips-triangle integrand is missing a V_0 factor (we’d expect *two* V_0 -shifted propagators in the loop, not one, since two of the three loop legs carry the excitation in the cyclic walk), or path-independence is a special property of the $n=2$ topology and fails generically. §5 pins this down.

5 Verification harness (what we ran before sending this)

To avoid embarrassing both of us, we ran four independent checks on every formula in this note. The script `verify_diagrammatics.py` (output in `verify_log.txt`) implements all four; this section summarises what was tested and what came back.

(A) Walk enumeration vs Laplacian (combinatorial half). For $n \in \{2, 3\}$ and $m \in \{0, \dots, 4\}$ we enumerate by brute force every sequence of (stay/flip, partner-rail) consistent with the strict rule “one of i, j carries the excitation at vertex v_{ij} ,” subject to start and end at rail 0. Sum the symbolic weights ($+v_{ij}$ for stay, $-v_{ij}$ for flip) and compare to $(L^m)_{00}$ from the matrix Laplacian (1). They agree *symbolically* at every (n, m) tested. So (1) is the correct symbol-level answer for every walk that satisfies the strict rule.

(B) Diagram reconstruction (atlas check). For each diagram of §2 we assemble the value Γ from your rules independently — excitation path, propagator V_0 -shift on every blue segment, $\pm v_{ij}$ at every vertex, explicit kernel from the topology — and compare to your stated formula. Five of six match exactly:

- $n=2, m=1$ stay — match.
- $n=2, m=2$ stay/stay — match (symbol *and* kernel).
- $n=2, m=2$ flip/flip — match (symbol *and* kernel).
- $n=3, m=2$ no-loop — match.
- $n=3, m=3$ stays-only triangle — symbol matches; kernel $(-i\omega + V_0 + 2r)^{-2}$ is independently verified by SymPy residue (test C).
- $n=3, m=3$ cyclic-flip triangle — **fails on the kernel**. The integrand you wrote evaluates to $[(-i\omega + 2r)(-i\omega + V_0 + 2r)]^{-1}$, not $(-i\omega + V_0 + 2r)^{-2}$. (See test C.)

One open question (sign discrepancy on the first $n=3, m=3$ diagram). The first $n=3, m=3$ diagram you draw has stated weight $+v_{01}v_{12}v_{20}$. Under the strict-rule walk enumeration, no $0 \rightarrow 0$ walk of length 3 that uses each of the three triangle edges exactly once has positive sign — the only candidates are the two cyclic walks $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ and $0 \rightarrow 2 \rightarrow 1 \rightarrow 0$, both with three flips and sign -1 each. So either (i) the diagram allows a bath–bath spectator vertex at V_2 , in which case (1) is incomplete and we need a larger formalism; or (ii) the stated $+$ is a slip and should be a $-$, in which case the two diagrams together give $-2 v_{01}v_{12}v_{20}$ matching the Laplacian coefficient $[v_{01}v_{12}v_{20}] (L^3)_{00} = -2$ which we also verified.

(C) Independent residue computation of every kernel. SymPy closes each kernel integrand in the upper half plane and evaluates the residues. Four kernels checked:

1. $\int (-i(\omega + \omega') + V_0 + r)^{-1} (i\omega' + r)^{-1} d\omega' / (2\pi) = (-i\omega + V_0 + 2r)^{-1}$. Match.
2. $\int (-i(\omega + \omega') + r)^{-1} (i\omega' + V_0 + r)^{-1} d\omega' / (2\pi) = (-i\omega + V_0 + 2r)^{-1}$. Match.
3. $\int (i\omega' + r)^{-2} (-i(\omega + \omega') + V_0 + r)^{-1} d\omega' / (2\pi) = (-i\omega + V_0 + 2r)^{-2}$. Match.
4. $\int (i\omega' + V_0 + r)^{-1} (i\omega' + r)^{-1} (-i(\omega + \omega') + r)^{-1} d\omega' / (2\pi) = [(-i\omega + 2r)(-i\omega + V_0 + 2r)]^{-1}$. **Disagrees with your stated $(-i\omega + V_0 + 2r)^{-2}$.**

(D) Sanity limits.

- $V_0 \rightarrow 0$ collapses excitation/bare propagators identically. The $n=2, m=2$ stay and flip kernels both reduce to $(-i\omega + 2r)^{-1}$ — consistent.
- Uniform coupling: $(L^m)_{00} = \frac{n-1}{n} (nv)^m$ verified symbolically for $n \in \{2, 3, 4\}$, $m \in \{1, \dots, 4\}$.

Verdict. The combinatorial half (§2 and §4) is solid: walk enumeration matches $(L^m)_{00}$ at every tested (n, m) , all sanity limits hold. The atlas reconstruction is solid for the $n=2$ rows and the $n=3, m=2$ no-loop case. Two independent issues survive: **(B-i)** the first $n=3, m=3$ diagram with sign $+$ has no walk-level explanation under the strict rule, and **(B-ii)** the second $n=3, m=3$ diagram's stated kernel is internally inconsistent with its own integrand. These are the things we'd ask you about before treating the corrected picture as settled.

6 Your four asks, re-cast through the corrected picture

(1) Generating function of the diagrams. Re-derived in two lines from the corrected rules:

- Strip the kernels. Each vertex carries $\pm v_{ij}$ (sign by stay/flip), and a length- m rail walk $0 \rightarrow 0$ accrues $\prod_{t=1}^m L_{a_{t-1}a_t}$ where a_t is the rail of the excitation just after vertex t . Sum over walks gives $(L^m)_{00}$.

- Generating function:

$$\mathcal{G}^{(n)}(z; \{v_{ij}\}) := \sum_{m \geq 0} (L^m)_{00} z^m = [(I - zL)^{-1}]_{00} = \frac{\det(I - zL)_{[0,0]}}{\det(I - zL)}.$$

Specialised to uniform v : pole at $z=1/(nv)$, zero at $z=1/v$, giving $\frac{n-1}{n}(nv)^m$. Per-channel structure is read off by keeping v_{ij} symbolic and expanding the cofactor.

(2) Atlas of loop diagrams — and yes, they are essentially trivial at fixed topology.

The corrected lesson is: the **loop topology** is a labelled multigraph Λ on the rail set, with one node per vertex and one edge per rail-segment between vertices; the excitation worldline is a distinguished tree inside Λ . Two facts:

- The kinematic kernel $K(\Lambda)$ depends only on Λ as an unlabelled multigraph (which leg is “the blue one” shifts by a contour translation, as in the two $n=2, m=2$ diagrams above).
- For the all-uniform- r propagator, $K(\Lambda)$ collapses to a product of single-loop integrals, one per independent cycle of Λ ; each independent cycle of length ℓ evaluates to $(-i\omega + V_0 + \ell r)^{-1}$ raised to a power that counts how many times that cycle sits on the worldline. The atlas of Λ ’s up to order m is finite and small:

m	loop topologies (independent cycles, no on-rail dressing)
0	tree (no loop)
1	tree (the off-rail receives one insertion, no closed cycle yet)
2	single 2-cycle (bubble) or tree (different rails)
3	single 3-cycle (triangle) or bubble + extra insertion or tree
4	4-cycle, two bubbles, bubble+triangle, single-cycle dressings, tree

A clean way to count: the loop topologies at order m are the connected multigraphs on $\leq m+1$ nodes with cyclomatic number $\leq m$ and edge multiset constrained by “one blue path through.” I’ll write the generating series for these once you confirm this is the right object.

(3) Asymptotics with v_{ij} unspecified. The Laplacian framing gives rather a lot for free:

- Let $0 = \lambda_0 < \lambda_1 \leq \dots \leq \lambda_{n-1}$ be the eigenvalues of L (Perron–Frobenius: simple top eigenvalue under any positive weights). Then $W_m^{(n)} \sim |\psi_{n-1}(0)|^2 \lambda_{n-1}^m$ as $m \rightarrow \infty$, with ψ_{n-1} the top eigenvector.
- For uniform v , $\lambda_{n-1} = nv$ with multiplicity $n-1$ and $|\psi_{n-1}(0)|^2$ collapses to $(n-1)/n$.
- For non-uniform v_{ij} , λ_{n-1} is bounded by Cauchy interlacing: $\max_a \sum_j v_{aj} \leq \lambda_{n-1} \leq 2 \max_a \sum_j v_{aj}$. The lower bound is tight for star graphs; the upper bound for regular ones.
- Subleading corrections come from the spectral gap $\lambda_{n-1} - \lambda_{n-2}$, which for graphs with no global symmetry is generically of order one. So one expects the standard $\lambda_{n-1}^m (1 + \mathcal{O}(\rho^m))$ pattern with $\rho = \lambda_{n-2}/\lambda_{n-1} < 1$.

(4) Statistics of v_{ij} — how do correlators enter? Yes, this is clean once stated as: the diagram is a polynomial in the v_{ij} , so disorder averages just slot in moment-by-moment.

A loop weight, ignoring signs and the path-of-the-excitation, is exactly the cycle product you wrote:

$$\prod_{\text{cycle}} v_{ab} v_{bc} \dots v_{za}.$$

At the disorder-averaged level,

$$\langle W_m^{(n)} \rangle = \sum_{w \in \text{walks}_m(0 \rightarrow 0)} \langle \prod_{e \in w} \sigma_e v_e \rangle = \langle (L^m)_{00} \rangle,$$

where $\sigma_e \in \{\pm 1\}$ is the local stay/flip sign. If the v_{ij} are i.i.d. with moments $\mu_k = \langle v^k \rangle$, then each walk contributes $\prod_e \sigma_e \mu_{c(e)}$ with $c(e)$ the multiplicity of edge e in the walk. The natural object for averaged asymptotics is therefore the *averaged Laplacian power* $\langle L^m \rangle$, and at $m \rightarrow \infty$ its top eigenvalue controls the rate.

If correlators of triple products $\langle v_{ab} v_{bc} v_{ca} \rangle$ are specified (i.e. a triangle's edges are correlated), they enter the $m \geq 3$ averaged sum exactly through walks that visit those three edges; the resulting series is still rational in z , with poles shifted by the disorder moments. I'd be happy to work out the "Wigner-style" averaged $\mathcal{G}^{(n)}$ in a follow-up if you want, once the loop atlas of (2) is settled.

7 One small worry about your note

You write: "If I assume [the ω -integration is path-independent] is always the case, then obviously odd- m diagrams cancel pairwise." I don't think this follows. The reason:

- At $n=2$, every walk $0 \rightarrow 0$ has an *even* number of flip vertices (rail-state parity). So the per-walk sign is $(-1)^{\text{flips}} = +1$ for *every* contributing walk, regardless of m . Your blanket per-diagram weight $(-v_{01})^m$ is therefore correct for m even, but for m odd the actual weight is $+v_{01}^m$, not $-v_{01}^m$. There's no cancellation to take advantage of: $(L^m)_{00} = 2^{m-1} v_{01}^m \neq 0$ for both parities.
- At $n > 2$, walks can have an odd flip count (e.g. $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ at $n=3, m=3$, three flips, sign -1). But these don't pair-cancel against anything: the no-flip companion has the same kernel and a $+$ sign, but they sit on different rail-cycle weights $v_{01} v_{12} v_{20}$ vs. $(v_{01} + v_{02})^3 + \dots$, so they don't cancel as algebraic objects. The Laplacian sums them with the correct relative signs and the result is generically nonzero.

I suspect the slip is in the per-diagram formula $(-i\omega + V_0 + r)^{-1} (-v_{01})^m (-i\omega + V_0 + r)^{-1} (-i\omega + V_0 + 2r)^{-(m-1)}$: the sign on v_{01} shouldn't be a flat $(-)$ because the per-walk sign depends on parity of flips, not on m . The corrected per-walk weight is $(+1)^{\text{stays}} (-1)^{\text{flips}} v_{01}^m$, and when summed over the 2^{m-1} even-flip walks at $n=2$ all signs are positive. That gives $W_m^{(n=2)} = 2^{m-1} v_{01}^m = \frac{1}{2} (2v_{01})^m$ for all $m \geq 1$, matching the Laplacian and your original $n=2$ formula.

8 Summary

- I dropped V_0 from the excitation-rail propagator, and I mis-counted loop topology for swap-and-return. Both errors are local to §3 of the 7 May note.
- The combinatorial half (§2 & §4) is independent of the kernel error and survives unchanged, including the $\frac{n-1}{n} (nv)^m$ formula, the closed-form $\mathcal{G}^{(n)}(z) = (1-zv)/(1-znv)$ at uniform v , and the marking / Schur / continued-fraction toolkit.
- Once the kernel is repaired with your rules, every $n=2, m$ diagram has the same kernel $(-i\omega + V_0 + r)^{-2} (-i\omega + V_0 + 2r)^{-(m-1)}$, and the per-walk *sign* is $+v_{01}^m$ for all valid walks (because rail-parity forces an even number of flips). The 2^{m-1} count therefore appears with a uniform $+$ sign, and the total at $n=2$ is $2^{m-1} v_{01}^m \times (-i\omega + V_0 + r)^{-2} (-i\omega + V_0 + 2r)^{-(m-1)}$, both even and odd m .
- A clean restatement of (2) the loop atlas, (3) the asymptotics, and (4) how disorder correlators enter, can each be carried through with the corrected rules. I'd appreciate a sanity check on the loop-topology labelling I sketched in §6(2) before I produce the full table.